Project 3 ENE418, spring 2026

This Assignment focuses on the **solution of partial differential equations (PDEs)** in engineering applications. You will implement several numerical algorithms in Python, perform analysis, generate visualisations, and deliver your work in the form of a written report.

Expected Deliverables

- a written report (maximum 10 pages, excluding appendices)
- python code submitted as supplementary material (in the appendix)

You do not have to write any background information on the phenomena described by the differential equations in the tasks.

Problem description

You will simulate **ground heat transfer around boreholes** using the **2D advection–diffusion equation**, which accounts for:

- Heat diffusion through soil
- Advection due to groundwater flow
- Heat injection/extraction via constant or time-varying source terms

The governing PDE is:

$$\frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) + Q(t)$$

Where:

- $T(x, y, t)$: ground temperature at position (x, y) at time t [°C].
- (v_x, v_y) : groundwater velocity components in direction x and y, respectively [m/s].
- α : thermal diffusivity [m^2/s].
- $Q(t)$: source term (heat injection/extraction) at the boreholes [°C/s].

The simulation domain is a **60 m × 60 m** square with:

- constant groundwater velocity field (group-specific),
- constant heat source (Tasks 1–3)
- time-varying heat source (group-specific Tasks 4–5),
- one borehole (Tasks 1–4),
- four boreholes (Task 5).

Table 1. Parameters common to all groups

Parameter	Value
Domain size	$L_x = L_y = 60$ m
Grid resolution	$N_x = N_y = 200$
Cell size	$\Delta x = \Delta y = 0.3$ m
Time step	$\Delta t = 0.1$ s
Total simulation time	$t_{final} = 10\,000$ s
Thermal diffusivity	$\alpha = 5 \times 10^{-3}$ m²/s
Initial temperature	$T_0 = 10$ °C
Constant source term (Tasks 1–3)	$Q = 0.2$ °C /s
Boundary conditions	No flux $\frac{\partial T}{\partial n} = 0$
Single borehole location (Tasks 1– 4)	$(x, y)_{borehole} = (30, 30)$ m
Four boreholes locations (Task 5)	$(x, y)_{borehole\ 1} = (19.8, 19.8)$ m $(x, y)_{borehole\ 2} = (39.9, 19.8)$ m $(x, y)_{borehole\ 3} = (19.8, 39.9)$ m $(x, y)_{borehole\ 4} = (39.9, 39.9)$ m

Table 2. Velocity field and time-varying source term for each group

Group	Velocity Field [m/s]	Time-varying source term (Tasks 4–5) [°C/s]
1	$v_x = 0.0008y^{1.4}$ $v_y = 0.006(x - 30) + 0.06e^{-y/20}$	$Q(t) = 0.1 + 0.5 \sin\left(\frac{2\pi t}{2000}\right)$
2	$v_x = -0.0008y^{1.4}$ $v_y = 0.01(x - 30) - 0.06e^{-y/20}$	$Q(t) = \begin{cases} 0.02 & \text{if} \quad \text{mod}(t/2000) < 1000 \\ 0 & \text{if} \quad \text{mod}(t/2000) \geq 1000 \end{cases}$
3	$v_x = -0.02(y - 40)$ $v_y = 0.02(x - 40)$	$Q(t) = \begin{cases} 0.15 & \text{if} \quad \text{mod}(t/1500) < 200 \\ 0 & \text{if} \quad \text{mod}(t/1500) \geq 200 \end{cases}$
4	$v_x = 0.0008x^{1.4}$ $v_y = 0.006(y - 30) + 0.06e^{-x/20}$	$Q(t) = 0.15\,\text{randn}(t)$
5	$v_x = 0.02(x + 20) + 0.01$ $v_y = 0.02(y - 20)$	$Q(t) = 0.3\,\text{randn}(t)$
6	$v_x = -0.02(x - 20) + 0.01$ $v_y = -0.02(y - 20)$	$Q(t) = 0.1 + 0.5 \sin\left(\frac{2\pi t}{2000}\right) + 0.4 \sin\left(\frac{2\pi t}{500}\right)$

NOTE: the function $\text{mod}(\quad)$ stands for the remainder (i.e. rest of the division) and the function $\text{randn}(t)$ stands for a random number sampled at each time t from the standard normal distribution.

TASK 1 - Pure Diffusion

Implement a finite-difference method to solve **2D pure diffusion equation** for a **single borehole** using the parameters provided in **Table 1**.
Assume the heat source *Q* is constant.

$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) + Q$$

Required outputs

- Plot temperature field at t = 2 000, 5 000, 10 000 s.
- Compare plume radius with the analytical diffusion length $L(t) = \sqrt{4\alpha t}$.
- Analyse how the solution changes when thermal diffusivity α is increased.
- Analyse how the solution changes when the source strength Q is increased or decreased (try to make an analogy with the behavior of a borehole between summer and winter time).

Bonus for Advanced Students

- Implement a numerical detection of plume radius: Identify the radial distance where $T - T_0$ falls below a threshold (e.g., 0.01 °C) and compare with $L(t)$.
- Grid refinement study: repeat simulations with (100×100, 200×200, 400×400). Comment on numerical anisotropy ("diamond plume") and convergence.
- Time-step convergence test: compare results using Δt = 0.1, 0.05, 0.02 s.

TASK 2 – Pure Advection

Implement a finite-difference method to solve the **2D pure advection equation** for a **single borehole** using the general parameters from Table 1 and **group-specific velocity field** in Table 2.
Assume the heat source *Q* is constant.

$$\frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} = Q$$

Required outputs

- Plot plume trajectory.
- Explain the directional behaviour of the plume based on your velocity field.
- Analyse the effect of increasing the time step Δt, and compare this effect with what you observed in Task 1.

Bonus for Advanced Students

- CFL analysis: compute and discuss numerical stability.
- Compare advection schemes (e.g. upwind vs. central).

TASK 3 – Advection + Diffusion

Implement a finite-difference method to solve the **full 2D advection-diffusion equation** for a **single borehole** using the general parameters in Table 1 and the group-specific velocity field in Table 2.
Assume the heat source *Q* is constant.

$$\frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) + Q$$

Required outputs

- Plot plume trajectory
- Explain the directional behaviour of the plume.

- Compare the results against Task 1 (diffusion) and Task 2 (advection) and discuss the competing roles of advection and diffusion.
- Analyse the effect of increasing the time step Δt and compare with your findings in Tasks 1 and 2.

Bonus for Advanced Students

- Evaluate Péclet number
$$\mathrm{Pe} = \frac{vL}{\alpha}$$
and discuss whether your system is advection- or diffusion-dominated.
- Energetics: compute the total "thermal mass"
$$M(t) = \iint (T - T_o)\, dxdy$$
and discuss conservation properties.

TASK 4 – Time-Varying Source

Extend the model from Task 3 to include a **time-varying source term $\boldsymbol{Q(t)}$**, as specified in Table 2 for each group.

Required outputs

- Plot temporal evolution of the temperature field.
- Discuss how the time variability $Q(t)$ affects the plume size, shape, direction and movement.
- Analyse the effect of increasing the time step Δt and compare this to the behaviour observed in Task 3.

Bonus for Advanced Students

- Track plume center $(x_c(t), y_c(t))$ over time.

TASK 5 – Multiple Boreholes

Extend the model from Task 4 to include **four boreholes**, located at the positions given in Table 1. Apply the **group-specific $\boldsymbol{Q(t)}$ independently at each borehole**.

Required outputs

- Plot the temperature field and visualise interference effects between the boreholes
- Discuss which boreholes interact most strongly, and explain why based on spacing, velocity field, and diffusion.

Bonus for Advanced Students

- Superposition test: compare the multiple-borehole simulation with the sum of four single-borehole simulations.

Note about the project:

This project is an academic exercise designed to illustrate numerical methods for solving partial differential equations. Several parameters (such as velocity magnitudes, source-term amplitudes, and diffusivity) have been intentionally chosen for pedagogical clarity rather than physical realism. Their values were selected to ensure that diffusion and advection effects are visually distinguishable within the simulation time and grid resolution provided. Additionally, all calculations are carried out using temperature expressed in degrees Celsius. For heat-transfer modelling, this is acceptable because the governing advection–diffusion equation depends only on temperature differences, which are identical in Celsius and Kelvin. However, in real-world engineering simulations involving thermodynamic properties or energy balances, absolute temperature in Kelvin would normally be required.